

ABHINAV SAXENA

Software Engineer/ Founding Engineer

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Education

The University of Sydney

Masters in Computer Science

Feb. 2026 – Present

Sydney, Australia

Bennett University

Bachelor of Technology in Computer Science and Engineering — CGPA: 8.99/10

Oct. 2020 – June 2024

Greater Noida, India

Experience

Esprit Analytique

June 2024 – Dec 2025

Lead Full Stack Developer

Remote

- Led the end-to-end development of **ProdigiDesk**, architecting a scalable full-stack system using **React** for the frontend and **Python** for the backend with **PostgreSQL**; drove multiple UI revamps to deliver smooth, responsive, and localized user experience using **JSON-based multilingual support**, and built the company's official website from scratch.
- Designed and implemented a robust application-wide **security layer** using **Python**, including **AES encryption** for secure API communication, a **hybrid AES + RSA key exchange mechanism**, and a **proxy-based API architecture**.
- Owned cloud deployment and DevOps strategy by containerizing applications with **Docker** and deploying frontend and backend services on **AWS EC2**; implemented **Azure CI/CD pipelines** for automated builds and deployments.

NEC Corporation India Pvt Ltd

Oct. 2023 – April 2024

Java Developer Intern

Noida, India

- Designed and developed **RESTful APIs** as a back-end **Java Spring Boot** developer, achieving **85% code coverage optimization**, delivering efficient, **scalable**, and robust functionality to support critical application components.
- Strengthened front-end performance by seamlessly integrating **REST APIs** with **responsive design** principles, resulting in a **25% increase** in user engagement and enhanced **cross-device compatibility**.
- Partnered with **cross-functional teams** to architect and implement innovative **UI designs**, prioritizing **user-centric features** that elevated user experience, streamlined interaction flows, and aligned with organizational goals for functionality and aesthetics.

Projects

GlassBox | *Python, Django REST Framework, PyTorch, scikit-learn, SHAP, scapy, React*

Aug 2026

- * Decoupled **ML pipeline** from serving: trained an **Isolation Forest** (200 trees) and a **PyTorch autoencoder** on benign-only CIC-IDS2018 traffic, serialized as joblib/.pt artifacts loaded once at startup via a singleton **ModelRegistry** so scoring never hits disk.
- * Ensured **honest evaluation**: thresholds tuned on a dedicated validation set with the held-out test set touched exactly once; autoencoder checkpointing selected on **validation PR-AUC** rather than reconstruction loss to avoid inflated benign-only metrics.
- * Wired **SHAP explainability** into the live alert path (**TreeExplainer** and **KernelExplainer**), gated to fire only on flagged flows, with explanations persisted per-**Alert** for re-analysis if thresholds are later retuned.

Localeora | *React, TypeScript, Django REST Framework, Node.js, Babel, ts-morph*

Aug 2026

- * Architected an **AST-based localization platform** that extracts user-facing strings directly from React/TSX/HTML source using a **Node.js extraction engine (Babel)**, decoupled from a **Django REST API** owning orchestration.
- * Designed a two-engine pipeline separating read-only extraction from source-rewriting via **ts-morph**, with each edit re-validated against a content hash before being applied, preventing silent corruption of the codebase.
- * Implemented a **dry-run-to-apply** workflow with reviewable diffs, and a provider-agnostic translation layer (**Google Cloud Translation, Azure AI Translator**) using **Python asyncio** for concurrent batch jobs across 30+ languages.

Technical Skills

Programming & Frameworks: Java, Python, C++, Dart, JavaScript, HTML, CSS, LaTeX, React, Spring Boot, Django, Express, Flutter, TensorFlow, Tailwind, Bootstrap, Node

Cloud, DevOps & Tools: AWS, Azure, Docker, Firebase, Git, GitHub, Gitea, Razorpay

Databases & Security: MongoDB, MySQL, JWT, AES, RSA

Achievements

- * Secured **4th place** in the **Smart India Hackathon (SIH)**, showing strong problem-solving and collaborative development skills.
- * Authored and published a research paper on **Covid-19 Detection Using VGG-16**, applying deep learning techniques for medical image analysis.